

xSTUDIO - An Open Source Playback and Review Application

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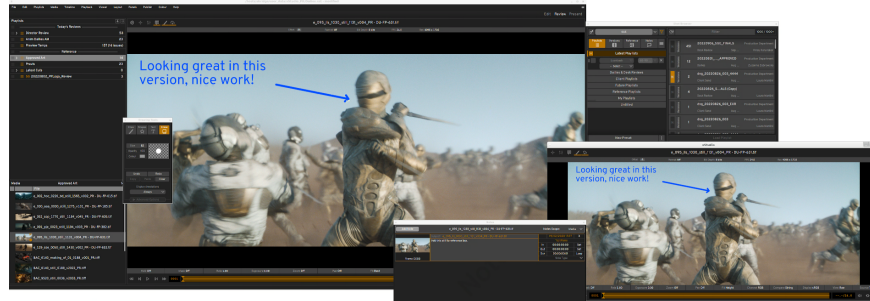


Figure 1: xSTUDIO's interface, showing notes, annotations and pipeline integration panels plus secondary viewer.

ABSTRACT

We present 'xSTUDIO', a high performance, easy to use playback and review application that is both an out-of-the-box solution, accessible to users at all levels of experience, and also a flexible platform that provides the core components for building new media player solutions where desired. The architecture, based on the 'actor model', ensures high concurrency for performance and flexibility as well as simplifying inter-thread and inter-process communications. Already in daily use by thousands of VFX and animation professionals at DNEG, xSTUDIO is now available to the community as an Open Source project.

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1 INTRODUCTION

Review and approval is a mission-critical, interactive action performed by multiple roles across varied systems, locations and platforms throughout the post production chain. As workflows become increasingly complex, fast paced and demanding with remote working and global collaboration being just two of several new challenges facing the industry, a modern and flexible Open Source review application that is accessible to users at all levels is keenly

desired by businesses and individuals. DNEG has been investing in the development of xSTUDIO for three years and the product was made Open Source in January 2023. xSTUDIO was built by Users for Users, and has evolved with live-production input from VFX Supervisors, Artists, Heads of CG, Producers, and others. xSTUDIO was born as a solution for the very real review and approval issues encountered today, and built in an open and adaptable framework for the issues of tomorrow. In this respect at least, it is unlike any solution we are aware of. In this talk we will give an overview of xSTUDIO, its unique design based on the 'actor model', its key features and performance. We will demonstrate how individuals and organisations can take advantage of it now, either deploying it as an out-of-the box solution or taking advantage of its flexibility to build their own novel playback solutions from xSTUDIO's core components.

2 PLAYBACK AND REVIEW CHALLENGES

2.1 Complex, Evolving Requirements

There are a wide variety of playback and review scenarios and workflows that are common to organisations across the film and TV post production sectors: from an individual artist quickly flipbooking on their latest playblast and interactively examining images up close to collaborative sequence based team reviews running across multiple sites. As the industry and associated technologies continue to evolve, such as the introduction of new high resolution and high frame rate video standards for example, so do these review requirements and the demands on the platforms that are intended to fulfil them. Workflows are also changing significantly as the joint challenges of increasing globalisation and remote working grow and it has been an ongoing challenge for studios to develop and maintain suitable review technology.

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2.2 xSTUDIO as a Solution

To overcome these significant challenges in the playback and review domain DNEG has invested in the development of a new, unified and flexible application that works as an out-of-the-box stand alone player whilst also allowing for more sophisticated, interactive review workflows and pipeline integrations. The modular design of the player also anticipates as yet undiscovered requirements by providing the flexibility to adapt and extend the application for new scenarios as they are identified. By making the project Open Source we hope that the community can benefit from xSTUDIO and contribute their own enhancements to the product so that it can meet an even broader set of requirements than the DNEG development team is already targeting.

3 XSTUDIO APPLICATION FEATURES

xSTUDIO is intended to be an intuitive, responsive and user friendly review solution with a shallow learning curve. By presenting a clear interface with design elements common to other popular playback applications our goal was that first time users could navigate the features and perform essential media playback and review activities quickly and easily. Advanced features for more complex and sophisticated review workflows, for example building and navigating collections of large media playlists, adding notes and on-screen annotations, flagging media with status indicators and so-on have been carefully designed and implemented.

C++ and Python APIs are provided for plugin and scripting development and, in particular, for pipeline integration. The API allows custom front ends for plugins to be created with Qt/QML allowing total flexibility for studios and individuals to build their own review platform with xSTUDIO.

The application has been designed with a modular approach where key components such as the media readers, viewport, overlays and color management themselves are implemented as plugin extensions that can be customised, extended or completely replaced without need to modify the core codebase. Careful optimisation of the playback engine should ensure that the most demanding video formats can be played, depending on the given system specification.

Accurate color management is guaranteed as xSTUDIO's color plugin integrates OpenColorIO into the viewport display graphics, executing transformations on the GPU at floating point precision. OCIO transforms are driven by media metadata values that can be set through simple API hooks.

4 XSTUDIO ARCHITECTURE

To deliver a review tool that includes a high performance core media player but that also maintains interactivity and responsiveness in the interface at all times requires a suitable architecture (See figure 2). This design should maximise concurrency when required and ensure that interdependent components are nevertheless non-blocking to their counterparties.

4.1 Core Components and Interaction

We note that a review solution that fulfils these goals requires a number of components that must interact effectively in an asynchronous manner to deliver the best experience for the user, including frame accurate playback, tight audio sync and an interface that

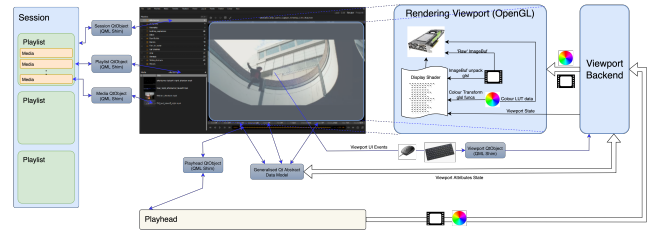


Figure 2: Schematic showing top level interactions between xSTUDIO's backend and UI layer

is always responsive. Such components might include the following: video display, audio output, core session data model (media, playlists and so-on), filesystem IO, video decode, video frame cache, python interpreter, GUI layer and event loop.

4.2 Employing the 'Actor Model'

With a traditional object oriented model we anticipated that managing queues of job requests and handling communication between these various processes using mutex and semaphore objects could become difficult to design, hard to maintain and perhaps difficult to optimise for best performance. Instead the application has been designed using the 'Actor Model'[Charousset et al. 2013] by employing the 'C++ Actor Framework' (CAF), an open source implementation of an actor model system, as the basis for the architecture. To our knowledge this is a novel approach for such a product.

4.3 Benefits of the Actor Model

With this approach the core application classes and plugins are implemented as 'actors' that interface only by sending, receiving and responding to messages. The CAF manages the message queue and executes actors' message handlers across a threadpool in a thread safe manner. This has the benefit of maximising concurrency, should it be required, whilst eradicating the need for direct implementation of thread locking that would otherwise be needed. The CAF and the actor model present further advantages for our application. A decentralised design with no main event loop maximizes flexibility: it is quite simple to instance multiple playheads and viewers to play back from a single media source actor. This opens the possibility for many player configurations with independent viewer windows, for example. This also facilitates a clear separation of the UI and backend components of the playback engine, viewer and interface, where the GUI event loop does not get interrupted by, nor will it interrupt, the core messaging system. The messaging system also provides a mechanism for serialising message data and passing it through network sockets allowing separate processes to connect transparently. This eliminates the need to implement an inter process communication layer which would otherwise be a significant challenge.

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